Niklas Medinger

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Education

exp. 2027 - 05/2022 **Ph.D. in Computer Science**, *Saarland University*, Saarbrücken, Graduate School of Computer Science; research at CISPA.

04/2022 - 10/2020 **Graduate School of Computer Science**, *Saarland University*, Saarbrücken, Prepatory Phase.

09/2020 - 10/2017 **B.Sc. Cybersecurity**, *Saarland University*, Saarbrücken.
Grade: 1.3

Employment

05/2026 - 05/2022 **Ph.D. Candidate**, *CISPA - Helmholtz Center for Information Security*, Saarbrücken.
Ph.D. candidate in the group of Cas Cremers; dissertation submission expected 11/2026. Research on the verification of cryptographic protocols in the symbolic model, focusing on increasing the expressiveness and automation of the Tamarin Prover and its application to real-world protocols.

11/2023 - 08/2023 **Applied Science Intern**, *Amazon Web Services Inc.*, Santa Clara.
Automated Reasoning Group. Supervised by Dimitra Giannakopoulou and Zvonimir Rakamarić; development of internal formal reasoning tools.

09/2020 - 04/2019 **Research Assistant**, *CISPA - Helmholtz Center for Information Security*, Saarbrücken.
Formal analysis of the WPA2 Wi-Fi protocol with the Tamarin Prover in the group of Cas Cremers, resulting in a USENIX Security 2020 publication.

04/2019 - 05/2018 **Research Assistant**, *CISPA - Helmholtz Center for Information Security*, Saarbrücken.
Research assistant in the group of Michael Backes.

Publications

11/2026 **Less Effort, Shorter Proofs: Reinforcement Learning for Security Protocol Analysis in Tamarin**, *Matthias Cosler, Cas Cremers, Bernd Finkbeiner, Mohamed Ghanem, and Niklas Medinger*, To appear in: Proceedings of the 2026 on ACM SIGSAC Conference on Computer and Communications Security (CCS 26).

05/2025 **Impossibility Results for Post-Compromise Security in Real-World Communication Systems**, *Cas Cremers, Niklas Medinger, and Aurora Naska*, Appeared in: Proceedings of the 46th IEEE Symposium on Security and Privacy (S&P 25).

10/2024 **Keeping Up with the KEMs: Stronger Security Notions for KEMs and automated analysis of KEM-based protocols**, *Cas Cremers, Alexander Dax, and Niklas Medinger*, Appeared in: Proceedings of the 2024 on ACM SIGSAC Conference on Computer and Communications Security (CCS 24).

08/2020 **A Formal Analysis of IEEE 802.11's WPA2: Countering the Kracks Caused by Cracking the Counters**, *Cas Cremers, Benjamin Kiesl, and Niklas Medinger*, Appeared in: Proceedings of the 29th USENIX Security Symposium (USENIX Security 2020).

Teaching Experience

04/2025 - 10/2024 **Teaching Assistant**, *Saarland University*, Saarbrücken.
Teaching assistant for the lecture *Formal Analysis of Real-World Security Protocols*.

06/2024 **Instructor**, *Summer School on Real-World Crypto and Privacy*, Vodice, Croatia.
Instructor for a hands-on tutorial on protocol verification with the Tamarin Prover.

04/2024 - 04/2022 **Teaching Assistant**, *Saarland University*, Saarbrücken.
Teaching assistant for the Bachelor- and Master Seminar of CISPA at Saarland University.

10/2022 - 04/2022 **Teaching Assistant**, *Saarland University*, Saarbrücken.
Teaching assistant for the lecture *Formal Analysis of Real-World Security Protocols*, including curriculum development and the creation and grading of exercises, projects, and exams.

05/2020 - 09/2018 **Student Tutor**, *Saarland University*, Saarbrücken.
Tutor for the lectures *Programming II* and *Foundations of Cybersecurity I*; voluntary co-creator and co-organizer of the programming pre-course for *Programming II*; tutor and one-week lecturer on formal proof techniques for the *Math Pre-Course*.